COMP5152
Advanced Data Analytics
Introduction

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Lecturer

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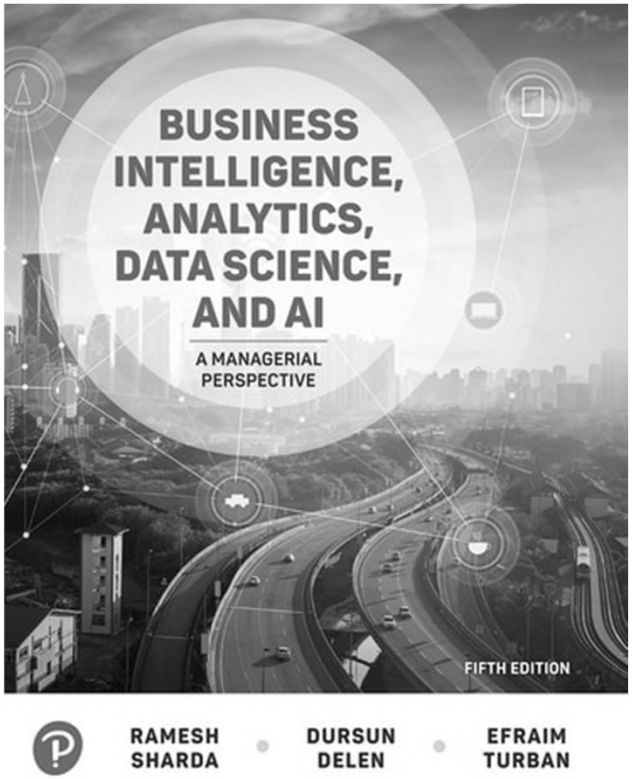
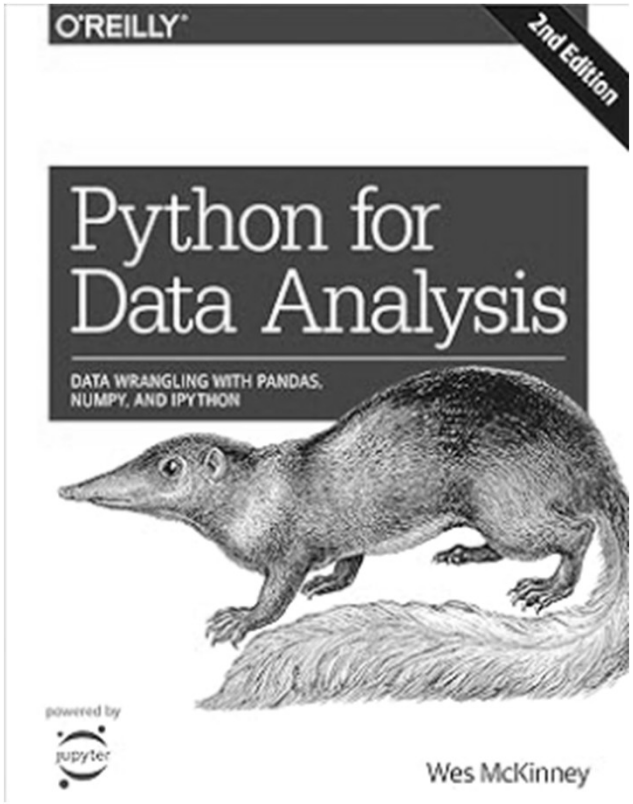
Readings

- Business Intelligence, Analytics, Data Science, and AI, 5th edition
- Published by Pearson (August 30, 2023) © 2024
- Authors : Ramesh Sharda , Efraim Turban



Readings

- WesMcKinney Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython 2nd Edition.
- Useful for your project



Assessment Scheme

- **Midterm Assessment(15%)**
  - Lecture 1 - 5
- **Group Project (15%)**
  - Report and Presentation at term end

- **Final Examination (70%)**
  - **3 hours** instead of 2 hours previously

Group Project

- 1-3 members/group
- Topic to work on
  - Prediction in a Selected Domain
  - Use of self-selected Public dataset or self-collected dataset
    - Clean up dataset to predict trends
  - Compare different methodologies: traditional vs machine learning
  - Write report and present the results/insight

- **The group members are given same score**
  - For any objection, email to subject lecturer by email waiman.pang@polyu.edu.hk with the justification and the signature of all group members

Course Topics and Schedule**(Tue class)**

	Date	***Topics (Tentative)***
1	13 Jan	Introduction to Data Analysis
2	20 Jan	Type of Data and Basic Statistics
3	27 Jan	Hypothesis testing and ANOVA
4	3 Feb	Linear Regression and Time Series Forecasting : ARMA and ARIMA
5	10 Feb	Feature Engineering and Dimensionality Reduction : PCA
	17 Feb	Chinese New Year Holiday
6	24 Feb	Classification: Logistics Regression, Decision Tree and Random Forest
7	3 Mar	Mid-term Exam
8	10 Mar	Clustering, Performance Measures and Deep Learning
9	17 Mar	Data sequence with Deep Learning
10	24 Mar	Association rule mining and Recommender system
11	31 Mar	Adv. analytics techniques: Bayesian Belief Network, Autoencoders, RBM, and GAN
12	7 Apr	Easter Holiday
13	14 Apr	Revision + Project Presentation

Course Topics and Schedule **(Thu class)**

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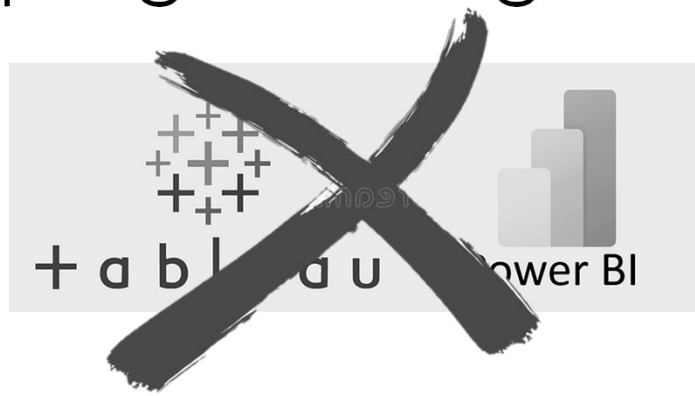
11	2 Apr	Adv. analytics techniques: Bayesian Belief Network, Autoencoders, RBM, and GAN
12	9 Apr	Project Presentation
13	16 Apr	Revision + Project Presentation

What is **NOT** expected in this course

- A certain domain specific data analytics
  - E.g. healthcare, finance, retail, and etc
- We learn the **<u>generic techniques</u>**
  - Come across selected cases
- Wide range of data analytics programming or tools
  - E.g. R, tableau, Power BI
- We focus on using **<u>Python</u>**



Why Data Analytics?

- Ways of **Solving Problem** or **Making Decision**
- E.g. How many stock we should prepare? Which PolyU restaurant we should go?

By Subjective Opinion

Describe problem to target person (friend, boss…)

↓

Collect Suggestion from target person

↓

Action base on Suggestion

By Objective Data

Find data related to the problem

↓

Analyze the data to find insight and solution

↓

Action base on insight

Why Data Analytics?

- Business cycle times are extremely compressed; faster, and more informed.
- Managers need the right information at the right time and in the right place
  (core of the modern approaches to BI).
- Organizations are being driven to capture, understand, and harness their data to support decision making and improve business operations.
- Legislation/regulations require leaders to document their business processes and sign off on the validity of the information they report to stakeholders.

Data is the core in Data Analytics

- Data Analytics or BI rely heavily on huge amount of data
- **Data explosion problem**
  - Automated data collection tools and mature database technology lead to tremendous amounts of data stored in databases, data warehouses and other information repositories
- We are drowning in data, but starving for knowledge!

Data and Its Analysis in Decision Making

- The amount of data doubles every two years.
- Decision making process requires an organization to collect and analyze vast stores of data.
- Computer applications have moved from
  - transaction-processing and monitoring activities => to
  - problem analysis and solution applications.
- These activities are done with cloud-based technologies, mostly accessed through mobile devices.

A Case Study : The Power of Data

https://www.youtube.com/watch?v=NXEL5F4_aKA

Takeaways from this case study

- You can use data to make decisions
- The decisions ma be complex – with tradeoffs because of capacity – which is why you may need <u>to automate the decisions</u>
- The rules you use will evolve over time, be refined
- You can <u>measure the consequences of decisions</u> – good vs. bad
- You can tie decisions to business goals

System for Decision Making

Nowadays Enterprise always relies on

- use of a set of tools or comprehensive system, called Decision Support System (DSS)
- They commonly include data warehousing, data mining, online analytical processing (OLAP), dashboards, and
- the use of cloud-based systems for decision support

Characteristics of Decision Support System (DSS)

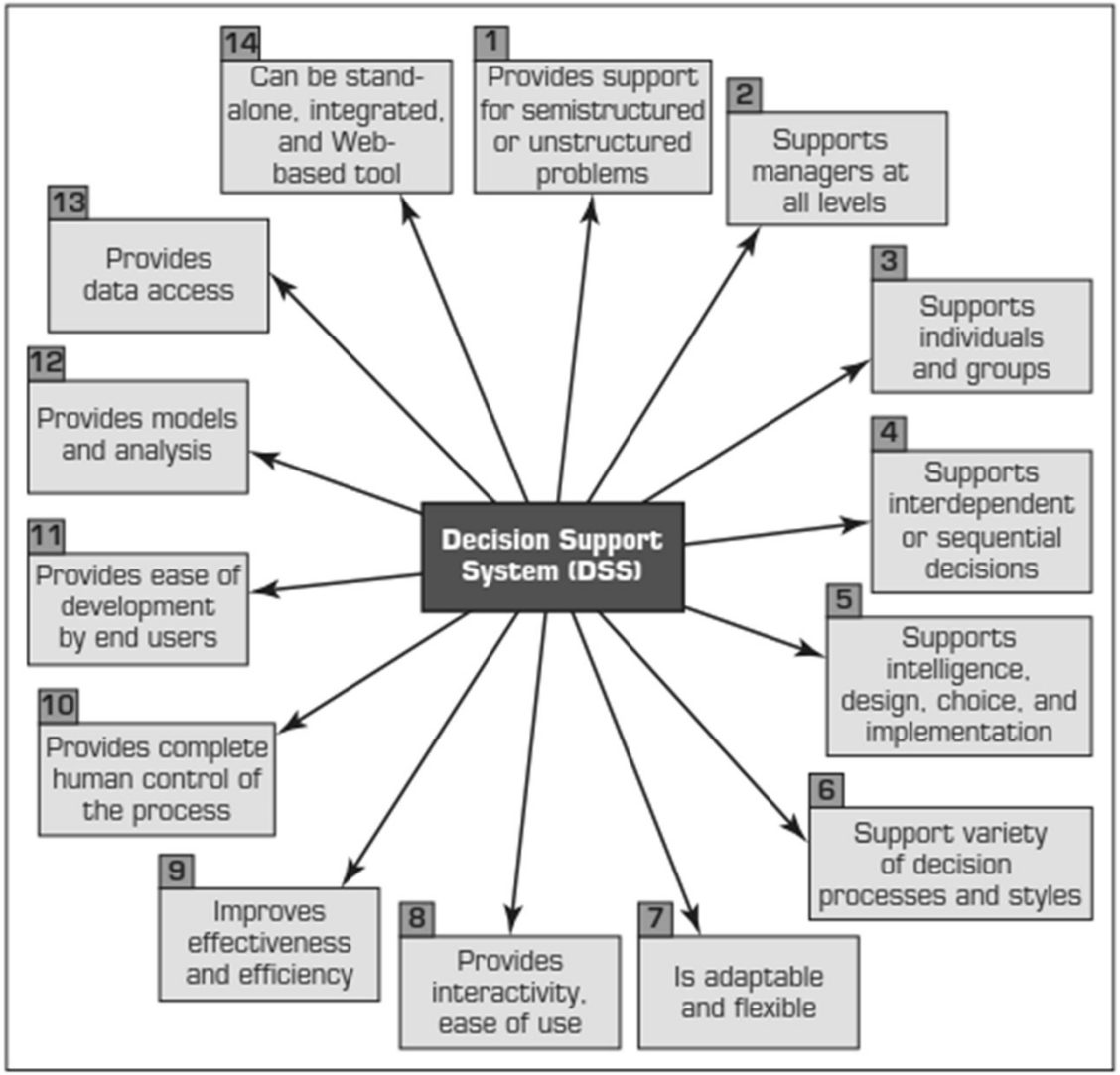


FIGURE 1.3 Key Characteristics and Capabilities of DSS.

Components of a Decision Support System

- A DSS can be composed of:

1. Data management subsystem
2. Model management subsystem
3. User interface subsystem
4. Knowledge-based management subsystem

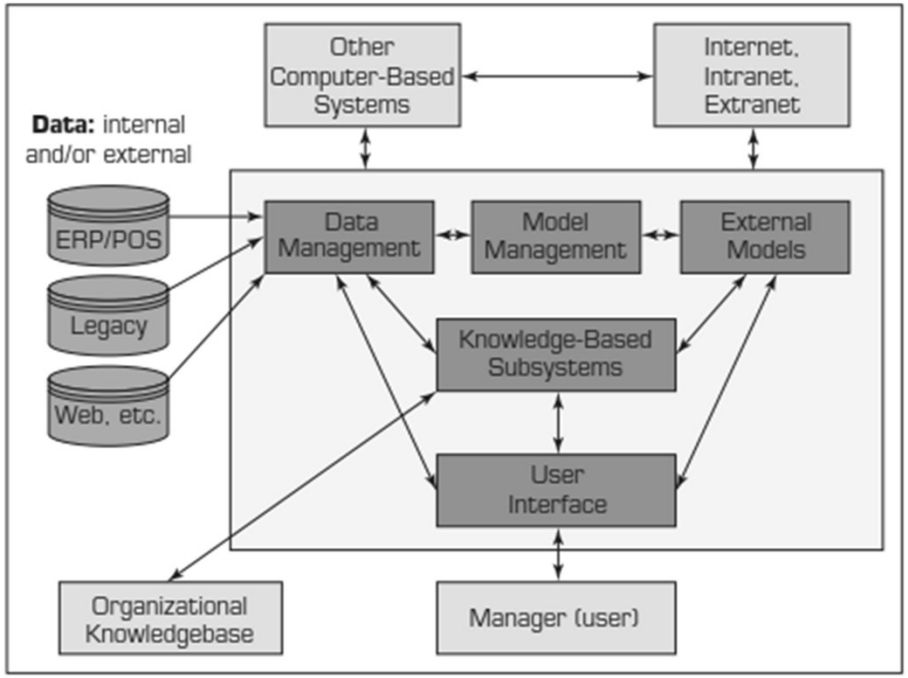


FIGURE 1.4 Schematic View of DSS.

1. Data Management Subsystem

- The data management subsystem is composed of the following elements:

1. DSS database
2. Database management system
3. Data directory
4. Query facility

- It can be interconnected with the corporate data warehouse (a repository for corporate relevant decision-making data)

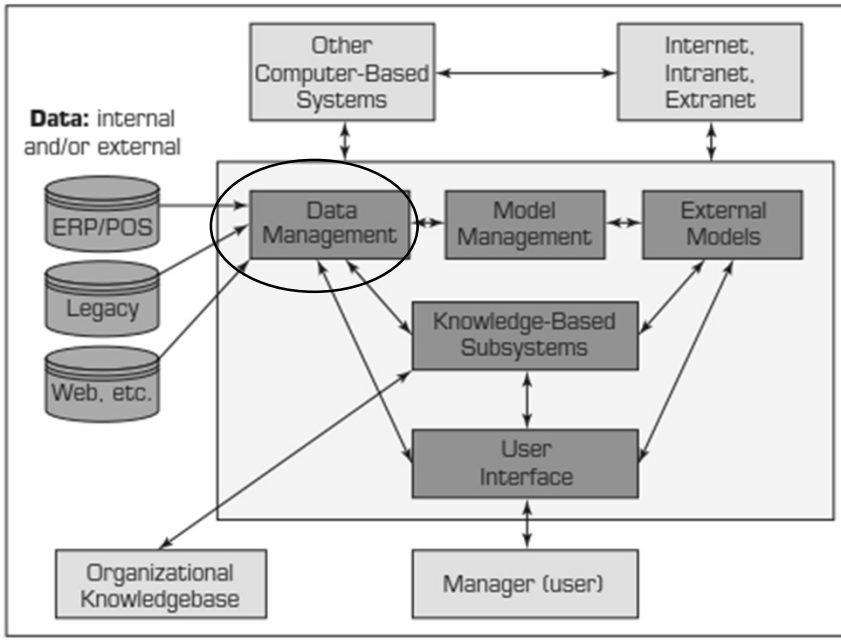


FIGURE 1.4 Schematic View of DSS.

2. Model Management Subsystem

- The model management subsystem is a component that includes quantitative models that provide the
  - system's analytical capabilities and
  - appropriate software management.

- The model management subsystem of a DSS is composed of the following elements:

1. Model base
2. MBMS (Model Base Management System)
3. Modeling language
4. Model directory
5. Model execution, integration, and command processor

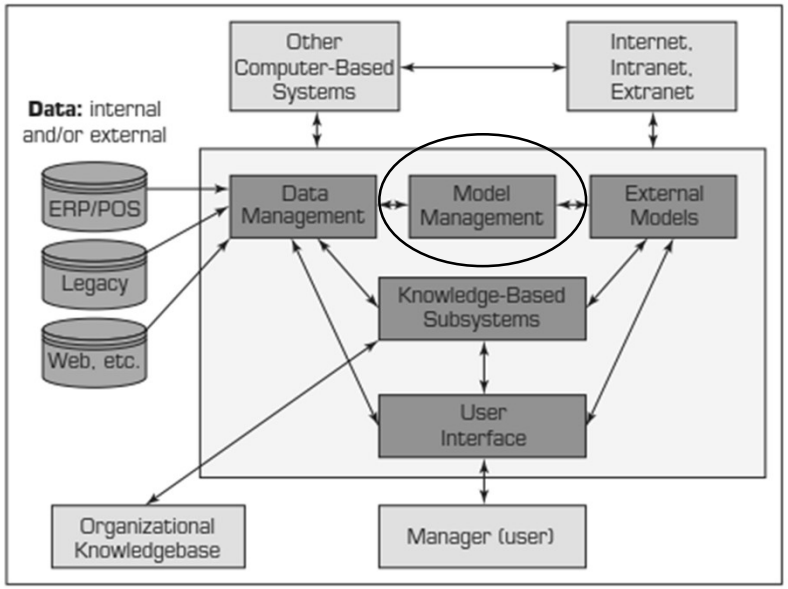


FIGURE 1.4 Schematic View of DSS.

3. The User Interface Subsystem

- The user is considered part of the system.
- The user communicates with and commands the DSS through the user interface subsystem.
- The Web browser has been recognized as an effective DSS GUI because it is
  - flexible,
  - user-friendly, and a
  - gateway to almost all sources of necessary information and data.
- Web browsers have led to the development of portals and dashboards (front end of many DSS).

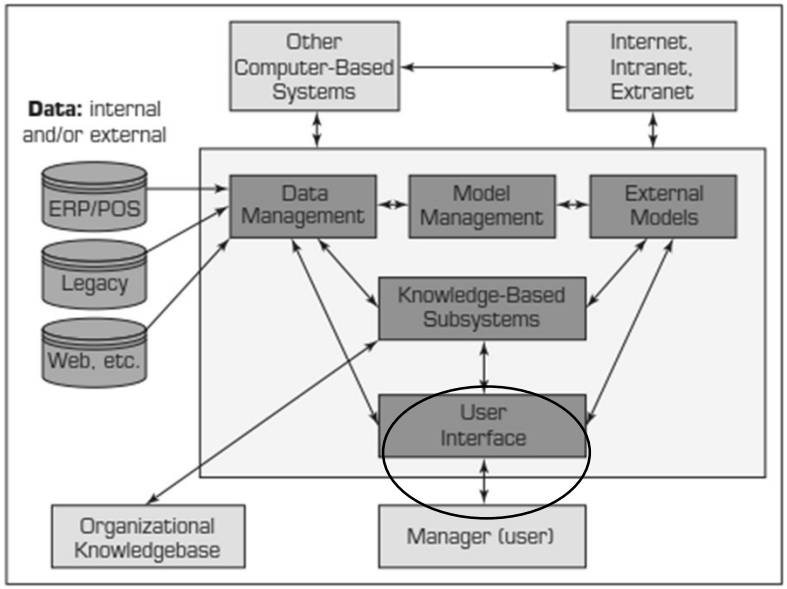


FIGURE 1.4 Schematic View of DSS.

4. The Knowledge-Based Management Subsystem

- The knowledge-based management subsystem provides intelligence
  - to augment the decision maker 's own intelligence
  - to help understand a user 's query and provide a consistent answer.
- Interconnected with the organization's knowledge repository or connected to thousands of external knowledge sources.
- User interface developments are closely tied to the major new advances in their knowledge-based systems.

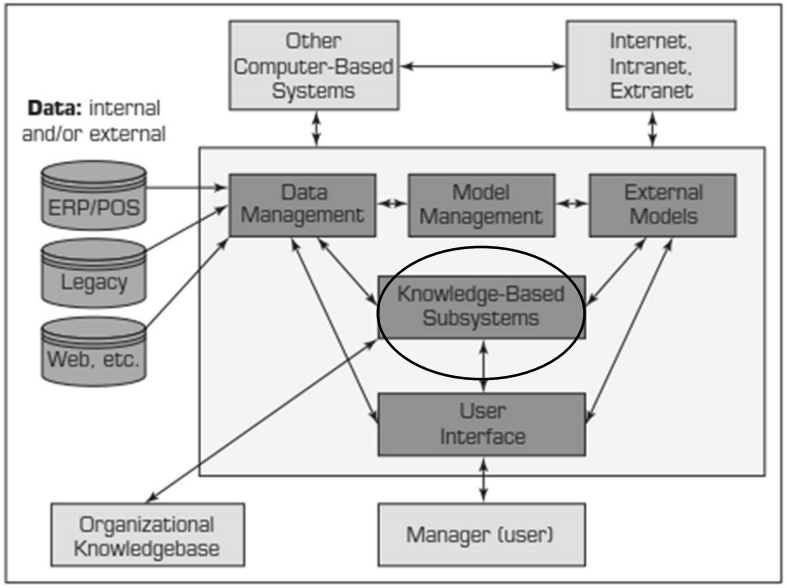


FIGURE 1.4 Schematic View of DSS.

Evolution Of Computerized Decision Support to Business Intelligence / Analytics / Data Science

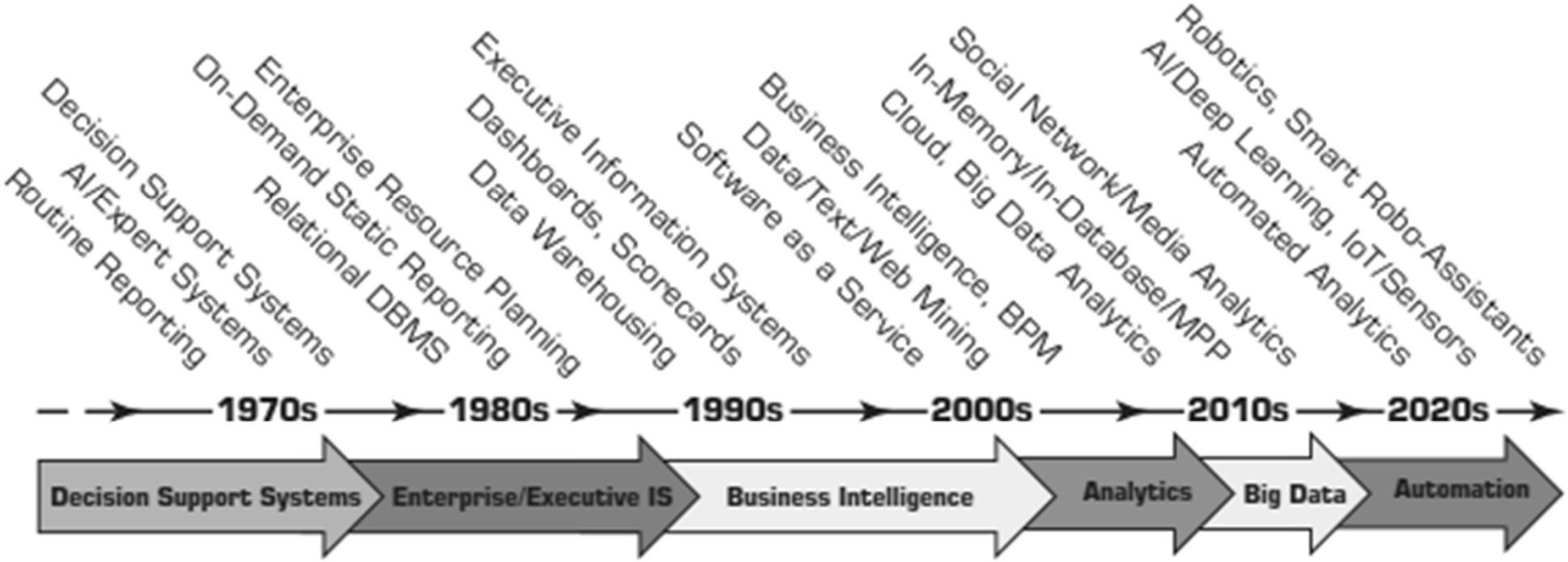


FIGURE 1.5 Evolution of Decision Support, Business Intelligence, Analytics, and AI.

Data Warehouse (DW) as a Foundation for BI

- BI systems rely on DWs as the information source for creating insight and supporting managerial decisions.
- DW is a subject-oriented, integrated, time-variant, nonvolatile collection of data in support of management 's decision-making process.
- The three main types of data sources for BI:
  - **Operational Data Stores (ODS):** provides a form of customer information file whose contents are updated throughout the course of business operations.
  - **Data Marts (DM):** a subset of a DW, typically consisting of a single subject area
  - **Enterprise Data Warehouses (EDW):** a large-scale data warehouse used across the enterprise for decision support.

Data Warehouse (DW) as a Foundation for BI (cont.)

- Data from many different sources can be extracted, transformed, and loaded into a DW for further access and analytics for decision support.
- Data is structured to be available in a form ready for analytical processing activities:
  - OLAP,
  - data mining,
  - querying,
  - reporting,
  - decision support applications

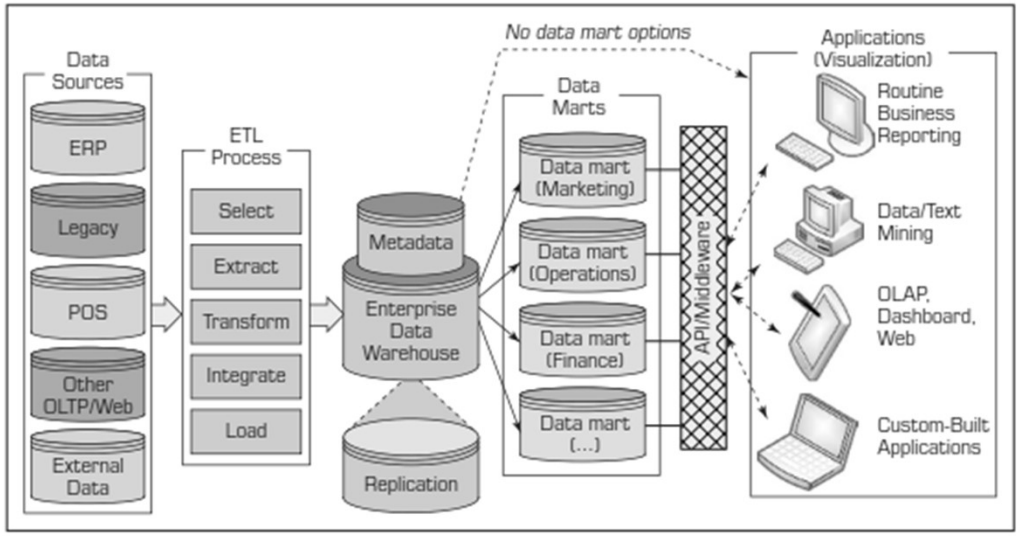
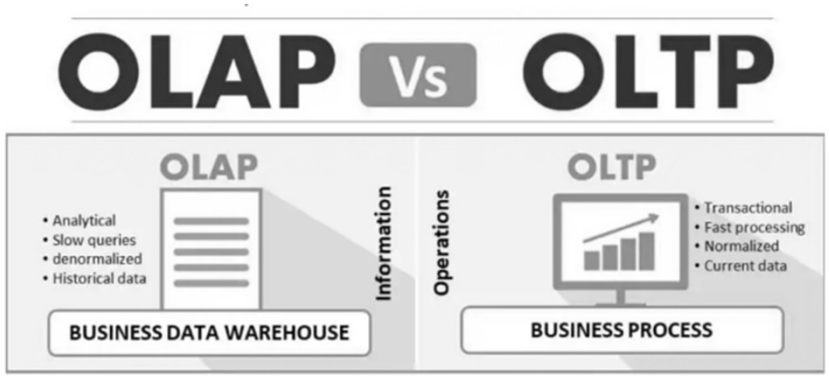


FIGURE 1.8 Data Warehouse Framework and Views.

Transaction Processing versus Analytic Processing

- Transaction processing systems are constantly involved in handling updates to what may be known as operational databases.
- OLTP (OnLine Transaction Processing) systems handle a company 's routine ongoing business, where the computer responds immediately to user requests.
- The OLTP system is efficient for transaction processing, but inefficient for end-user ad hoc reports, queries, and analysis.
- In contrast, a DW is typically a distinct system that provides storage for data that will be used for analysis by OLAP (OnLine Analytical Processing) systems.



Business Intelligence

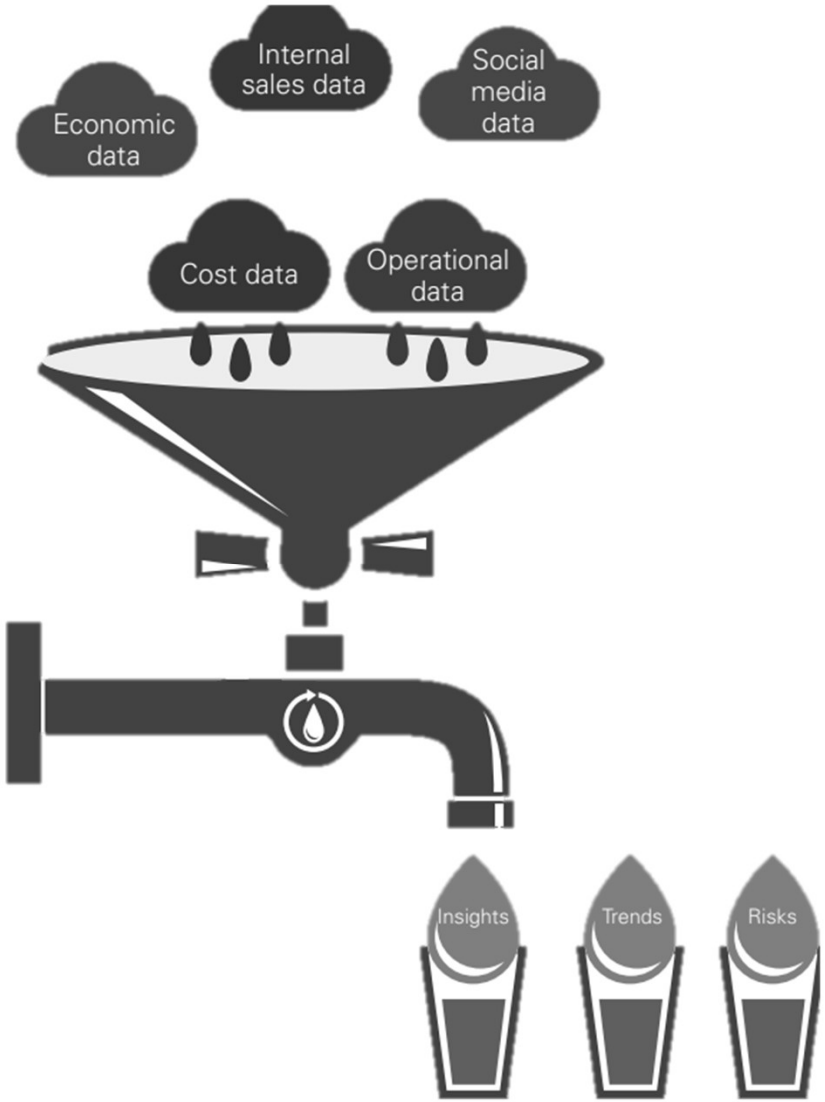
- The fundamental reasons for investing in BI must be aligned with the company's business strategy.
- BI improve company business processes and transforming it to be more data driven.
- BI tools adopts new technologies and sometimes integrated among themselves, resulting in six key trends:

1. Big Data
2. Focus on customer experience as opposed to just operational efficiency.
3. Mobile and even newer user interfaces—visual, voice, mobile.
4. Predictive and prescriptive analytics, machine learning, artificial intelligence.
5. Migration to cloud.
6. Much greater focus on security and privacy protection.

What is Data Analytics?

Data analytics is the process of examining data to **uncover useful information** and support **decision-making**.

The goal is to discover patterns, trends, and insights that would otherwise be hidden in the mass of data



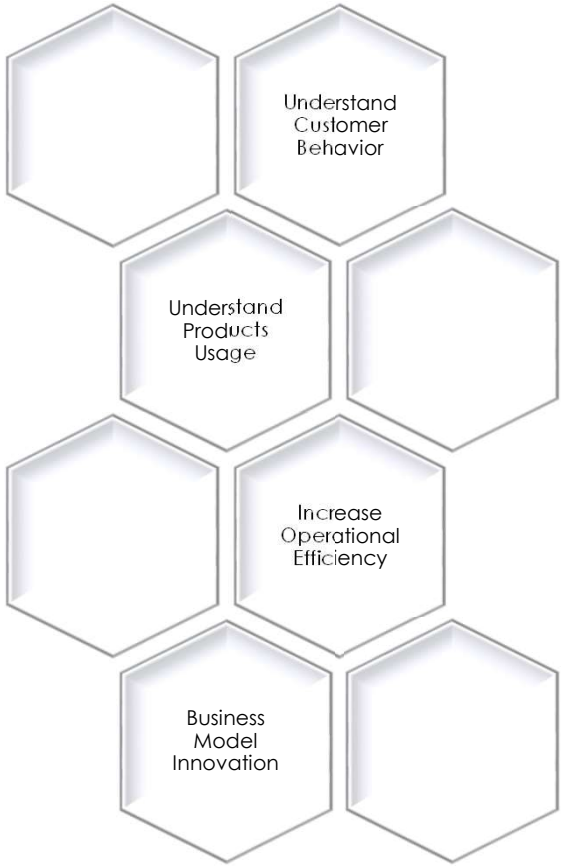
Source : KPMG

Who uses Data Analytics

- Data analytics has been adopted by nearly all sectors and industries
  - Banking And Securities
  - Media & Entertainment
  - Healthcare
  - Hotel & Hospitality
  - Manufacturing
  - Retail
- Large amounts of data
- Turnarounds are quick

Primary Focus for Analytics

- **Understand Customer Behavior**
  - To make a favor to customer in order to maximize profit
- **Understand Products Usage**
  - Improve new products by observing how it is used
- **Increase Operational Efficiency**
  - Reduce time and cost
- **Business Model Innovation**
  - Use data to create/support a new market



Data Analytics Life Cycle

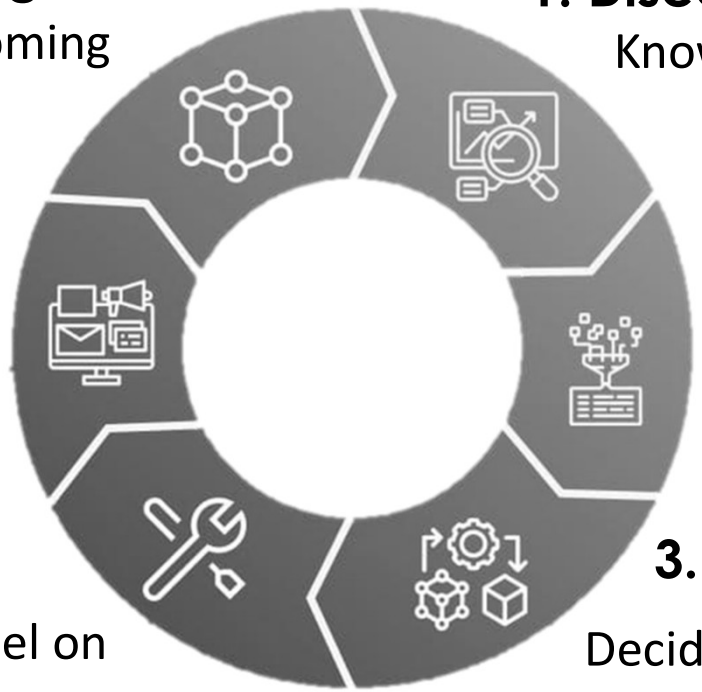
- We focus more on 3 and 4 in this course

6. Operationalizing

Deploy the model for the coming operational data

1. Discovery

Know what problem you are solving



5. Communicate results

Present your findings clearly

2. Data Preparation

Identify useful data, and ETL (extract, transform, load)

4. Build & Execute Model

Implement your decided model on the selected data

3. Model Planning

Decide which models (/solution/tools) to use

Data Analytics Pipeline

- It involves **collecting** raw data from different sources, **cleaning and organizing (ETL)** it, and using tools and techniques to **analyze** and **present** it.

Data Transformation Revisit

Source	Video views	Cost	Clicks	Conversions
All	5 914	2 642	1 925	1 104
	1 450	-	635	563
	2 321	1 240	897	356
	1 313	-	211	98
	830	284	182	87

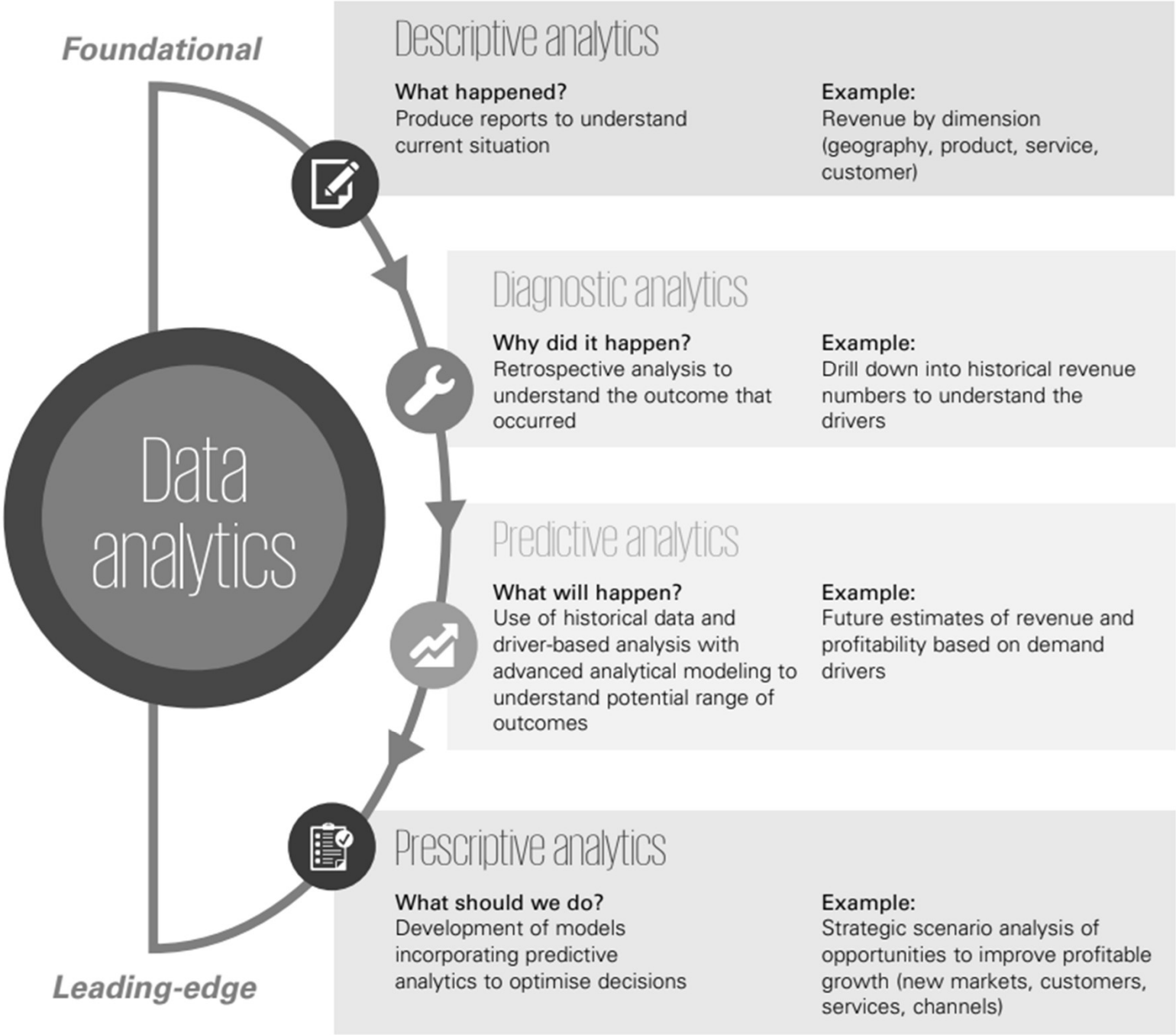
https://youtu.be/TY0NAPAWR3M

Types of Data Analytics

- Analytics not a tool/technology, rather it is the way of thinking and acting on data.

Descriptive Analytics	What has happened?
Diagnostic Analytics	Where & Why did it happen?
Predictive Analytics	What might happen?
Prescriptive Analytics	What should we do?

Types of Data Analysis



Source: KPMG

Descriptive analytics –What happened?

- insight into the past
- aims to provide insight into what has happened.
- There are various methods and technologies that are involved in descriptive analytics like frequency distribution, dashboards, business activity monitoring, content analytics, geospatial analytics.

Diagnostic analytics – Where & Why did it happen?

- to understand where and why
- Diagnostic analytics focuses on analysis of data to find out the causes of the event and relates to Root-cause analysis.
- The method/technologies include A/B testing, data mining and interactive visualization, pattern/anomaly detection and clustering/classification

Predictive analytics – What will happen?

- to understand the future
- Predictive analytics helps model and forecast what might happen.
- involves technologies like data mining, forecasting, clustering, machine learning and simulations.
- It tells "What is likelihood / probability of something happening in future?". In general predictive analytics is probabilistic in nature.

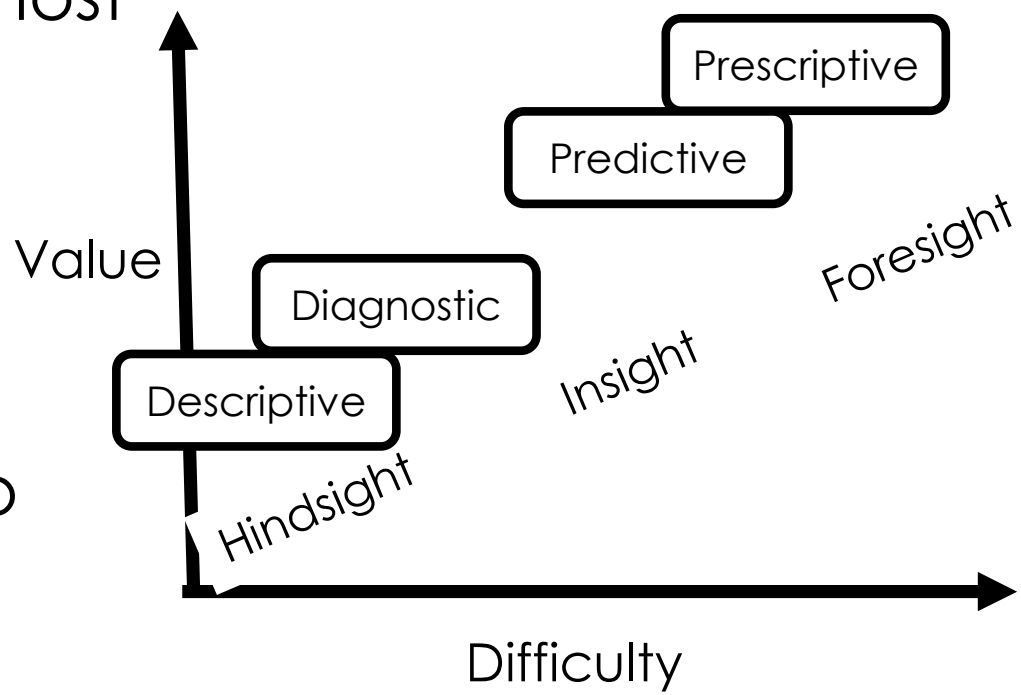
Prescriptive analytics – Make it happen

- to advice on possible outcomes
- seeks to determine the best solution or outcome among various choices, given the known parameters.
- The methods and technologies include fuzzy logic, optimization, expert system, and generative AI.

	Level of popularity	Answer to question…	Example Tools/ Techniques
Descriptive analytics	Used by almost all organizations	What happened?	Sampling, mean, mode, median, standard deviation, range, variance, histogram, quartiles, and frequency distributions
Diagnostic analytics	Used by many organizations	Where and/or Why exactly?	A/B Testing, Cluster and factor analysis, multiple regression, data mining
Predictive analytics	Used by a small but growing group of organizations	What are trends?	Cluster analysis, association analysis, multiple regression, decision tree, neural networks, simulation, and forecasting tools
Prescriptive analytics	Not yet widespread	What to do?	Fuzzy logic, Expert system, optimization, generative AI

Comparison of the 4 types of analytics

- Descriptive analysis is the most common and easiest
  - Understand distribution, variance
- **When to use** which type depends on
  - The problem you want to solve
  - What data do you have
  - Any assumption you can put to your model



Healthcare Analytics (Preventing Falls in Seniors)

- Accidental falls are a major health risk for adults age 65+ years. The costs of falls pose a significant strain on the healthcare system; estimated to be $34 billion in 2013 alone.
- Humana is a major health insurance company, with 3.2 million members, most of whom are seniors. Keeping its senior members well and helping them live safely is a key business objective. Fall prevention is an important component.

Healthcare Analytics (Preventing Falls in Seniors)

- As such, there is a need for an accurate method to identify individuals at greatest risk of falling so that they can be proactively managed for fall prevention.
- Administrative medical and pharmacy claims, clinical data, temporal clinical patterns, and other data is needed to identify individuals at high risk of falling over a time horizon.
- Today, the Falls PM (Predictive Models) is central to Humana's ability to identify seniors who could benefit from fall mitigation interventions. Using the PM, Humana was able to identify 20,000 consumers at a high risk of falling who benefited from this program. Identified consumers wear a device that detects falls and alerts a service for immediate assistance.

Retail Analytics

- Goal of the analytics industry is to use information about business suppliers, customers, employees, and stakeholders to make better decisions.
- Example:
  - Amazon's enormous investments in analytics to power their value chain.
  - Walmart & Target have invested millions of dollars in analytics for their supply chains.

Retail Analytics (cont.)

- Examples of key questions that can be answered through analytics, and the potential business value derived from fielding such analytics.

TABLE 1.1 Examples of Analytics Applications in the Retail Value Chain

Analytic Application	Business Question	Business Value
Inventory Optimization	1. Which products have high demand? 2. Which products are slow moving or becoming obsolete?	1. Forecast the consumption of fast-moving products and order them with sufficient inventory to avoid a stock out scenario. 2. Perform fast inventory turnover of slow-moving products by combining them with one in high demand.
Price Elasticity	1. How much net margin do I have on the product? 2. How much discount can I give on this product?	1. Markdown prices for each product can be optimized to reduce the margin dollar loss. 2. Optimized price for the bundle of products is identified to save the margin dollar.
Market-Basket Analysis	1. What products should I combine to create a bundle offer? 2. Should I combine products based on slow-moving and fast-moving characteristics? 3. Should I create a bundle from the same category or a different category line?	1. The affinity analysis identifies the hidden correlations between the products, which can help in following values: a. Strategize the product bundle offering based on focus on inventory or margin. b. Increase cross-selling or up-selling by creating bundle from different categories or the same categories, respectively.
Shopper Insight	1. Which customer is buying what product at what location?	1. By customer segmentation, the business owner can create personalized offers resulting in better customer experience and retention of the customer.
Customer Churn Analysis	1. Who are the customers who will not return? 2. How much business will I lose? 3. How can I retain the customers? 4. What demography of customer is my loyal customer?	1. Businesses can identify the customer and product relationships that are not working and show high churn. Thus, they can have better focus on product quality and the reason for that churn. 2. Based on the customer lifetime value (LTV), the business can do targeted marketing resulting in retention of the customer.

Retail Analytics (cont.)

TABLE 1.1 Examples of Analytics Applications in the Retail Value Chain

Analytic Application	Business Question	Business Value
Channel Analysis	1. Which channel has lower customer acquisition cost? 2. Which channel has better customer retention? 3. Which channel is more profitable?	1. Marketing budget can be optimized based on insight for better return on investment.
New Store Analysis	1. What location should I open? 2. What and how much opening inventory should I keep?	1. Best practices of other locations and channels can be used to get a jump-start. 2. Comparison with competitor data can help to create a differentiator to attract the new customers.
Store Layout	1. How should I do store layout for better topline? 2. How can I increase my in-store customer experience?	1. Understand the association of products to decide store layout and better alignment with customer needs. 2. Workforce deployment can be planned for better customer interactivity and thus satisfying customer experience.
Video Analytics	1. What demography is entering the store during the peak period of sales? 2. How can I identify a customer with high LTV at the store entrance so that a better personalized experience can be provided to this customer?	1. In-store promotions and events can be planned based on the demography of incoming traffic. 2. Targeted customer engagement and instant discount enhances the customer experience resulting in higher retention.

More Cases Studies

facebook

- Personalised recommendations
- Ad targeting
- Content optimisation

UBER

- Ride Matching
- Dynamic Pricing
- Route Optimization
- Predictive Maintenance

AXA

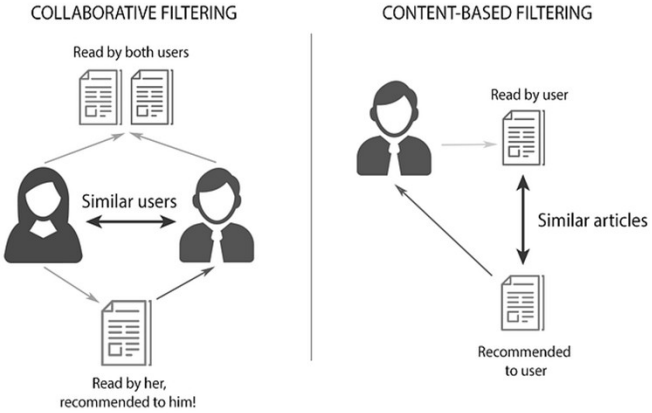
- Matching with Financial Consultant
- Discovery of correlation between claims risk & energy consumption

HKTV mall

- Inventory forecasting
- CRM
- Product Recommendation and Review Dashboard

Facebook

- Well known social media platform
- **Personalised recommendations**
  - Facebook uses data analytics to recommend content to users based on their interests, past interactions, and demographic information



Source : Decoding Your Posts: How Facebook Uses Data Analytics

Facebook (2)

- **Ad targeting**
  - Through data analysis, Facebook delivers targeted Advertisement to users based on their preferences, behaviors, and online activity.

- **Content optimisation**
  - By analysing user posts and interactions, Facebook optimises its platform to enhance user engagement, improve content visibility, and foster community interactions.

Source: Decoding Your Posts: How Facebook Uses Data Analytics

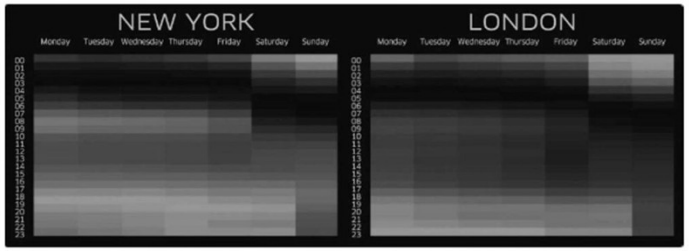
Uber

- Uber's system collect Big Data for predictive modeling including vehicles locations, traffic patterns, user demand, driver availability, and weather or events
- **Ride Matching and Allocation in Uber**
  - continuously processes data points such as current bookings, driver locations, optimal route paths, traffic updates, and rider preferences to match riders with the best available drivers

Source : Data Analytics Case Study on Optimizing Bookings for Uber
5 ways Uber is using AI [Case Study] [2024]

Uber (2)

- **Dynamic Pricing (Surge Pricing) in Uber**
  - Uber using machine-learning algorithms to determine the pricing based on analyzing various factors such as historical data, current supply and demand, time of day, weather conditions, and local events to set prices dynamically.



When Uber trips occur throughout the week in New York City and London. The brightness levels per hour and day are compared to the city itself. All times are standardized to the local time zone and expressed in military time (i.e. 20 is 20:00, or 8 pm).

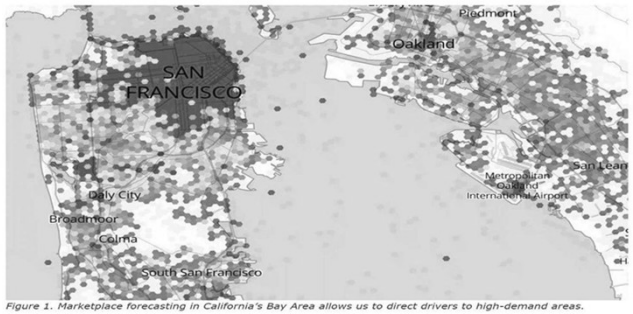


Figure 1. Marketplace forecasting in California's Bay Area allows us to direct drivers to high-demand areas.

5 ways Uber is using AI [Case Study] [2024]

Uber (3)

- **Route Optimization**
  - leverage predictive analytics to anticipate changes in traffic patterns and adjust routes proactively rather than reactively
  - historical traffic patterns, real-time speed data, and predictive models, to ensure that the suggested routes are both time-efficient and less prone to sudden delays.
- **Predictive Maintenance**
  - forecast vehicle issues before they become problematic
  - analyzing data from vehicle sensors and using machine learning models to predict when a vehicle component is likely to fail and proactively schedule maintenance.

5 ways Uber is using AI [Case Study] [2024]

AXA Hong Kong

- Use of AI to transform all areas of the business: marketing, operations, finance, etc.
- **engaging different teams** to use data and analytics effectively on a daily basis
- **Matching between customers and financial consultants**
  - leveraging big data to **analyzing their pattern of similarities** (e.g. age, family profile and residential address)
  - This initiative has successfully enhanced customer experience and generated better business results.

Source: AXA Hong Kong and Macau wins Big Data - Insurance category at HKB Technology Excellence Awards 2022

AXA Hong Kong (2)

GO GREEN home insurance campaign

- discovery of a correlation between claims risk and energy consumption
  - text mining to understand the home claim data
  - external energy consumption and weather data from the government's open database.
- tested the hypothesis that the **longer stay-at-home** due to COVID lead to **higher home energy consumption**, hence **higher claim frequency**, and weather factors affecting energy consumption.
- **a positive correlation** between home claims, energy consumption, and a few weather factors is mathematically proven
- Green Score for each individual household and leveraged **a tree-based machine learning** algorithm to indicate the "green" level of households

Source: AXA Hong Kong and Macau wins Big Data - Insurance category at HKB Technology Excellence Awards 2022

HKTVMall

- Uses data as a powerful tool for **inventory forecasting**
  - Especially during promotional periods where the sales increment could be well over 100 %!!
  - The pain point comes when retailers don't have enough stock
- **Customer Relationship Management (CRM)**
  - Because HKTVmall has so much information based on a customer's **browsing history and purchase behaviors**
  - companies are able to launch new products and easily target promotions at prospective buyers. This leads to higher **conversion rates**.

Source : Shopping Made Easy

HKTVMall (2)

- **Product Recommendation**
  - for add-on offering is a good example. After a customer makes a purchase, they are prompted to buy further products that might interest them.
  - Through this feature, the company has managed to increase conversion rates by up to 15 per cent, or HKD100 per basket.

HKTVMall – Product Review Dashboard

https://youtu.be/JobPgGj5-mc

Summary

- Overview of the basics of data analytics
- Life cycle, pipeline of data analytics
- 4 Major types of data analytics
  - Descriptive, Diagnostic, Predictive, Prescriptive
- Use case studies